

Mathematical Model of the Oboe Reed Full Crow

1 Virtual Fundamental

The virtual fundamental f_0 is the implied pitch from which the entire harmonic series descends:

$$f_0 = \frac{f_{C,\text{lowest}}}{2^k}$$

where $f_{C,\text{lowest}}$ is the lowest C detected in the spectrum and k is chosen such that $f_0 \approx C3$ (130.8 Hz).

In practice: if the lowest detected C is C5, use $k = 2$ (divide by 4); if C6, use $k = 3$ (divide by 8). Both resolve to the same $f_0 = C3$.

2 The Harmonic Series

All partials are integer multiples of the virtual fundamental:

$$f_n = n \cdot f_0, \quad n \in \mathbb{Z}^+$$

3 The Octave Set (“The C’s”)

The octave set $\mathcal{O}$ is the subset of harmonics where n is a power of 2:

$$\mathcal{O} = \{f_n : n = 2^j, j \in \mathbb{Z}_{\geq 0}\}$$

j	n	Frequency	Note
0	1	$f_0 = 130.8$ Hz	C3
1	2	$2f_0 = 261.6$ Hz	C4
2	4	$4f_0 = 523.3$ Hz	C5
3	8	$8f_0 = 1046.5$ Hz	C6
4	16	$16f_0 = 2093$ Hz	C7
5	32	$32f_0 = 4186$ Hz	C8

4 The Buzz Set

The buzz set $\mathcal{B}$ contains every harmonic *not* in $\mathcal{O}$:

$$\mathcal{B} = \{f_n : n \in \mathbb{Z}^+, n \neq 2^j \text{ for any } j\}$$

Grouped by octave band j , spanning $2^j f_0$ to $2^{j+1} f_0$:

$$\mathcal{B}_j = \{n \cdot f_0 : 2^j < n < 2^{j+1}\}$$

$$|\mathcal{B}_j| = 2^j - 1 \quad (\text{band density doubles each octave})$$

For the crow’s principal range:

Band	Octave span	n values	Count $ \mathcal{B}_j $	Notes lit up
$\mathcal{B}_3$	C6 $\rightarrow$ C7	9, 10, $\dots$, 15	7	D, E, F $\sharp$, G, A, B $\flat$, B
$\mathcal{B}_4$	C7 $\rightarrow$ C8	17, 18, $\dots$, 31	15	all chromatic tones + microtones

5 Richness Metric

A rich crow has high-amplitude components spread across $\mathcal{B}_3$ and $\mathcal{B}_4$. A thin or speaking reed concentrates energy at the octave boundaries ($n = 8, 16, 32$) with weak intermediate harmonics.

The **buzz fill ratio** per band quantifies this:

$$R_j = \frac{\#\{n \in \mathcal{B}_j : A_n > \alpha\}}{|\mathcal{B}_j|}$$

where A_n is the amplitude of harmonic n and α is a detection threshold.

$$R_j \rightarrow 1 \implies \text{full, rich buzz} \quad R_j \rightarrow 0 \implies \text{thin, octave-only crow}$$